Airfoil Inverse Design Optimization Report — Optimization_0074

Rate of Accuracy	Total Time Taken	Selected Base Model	Dataset Trust Degree
99.98%	**111.10 s**	**Airfoil_1426 (Try #4)**	**93.1%**

1. User Inputs & Optimization Configuration

Flight Regime: Re = 8.10M

Minimum t/c: 0.120

Primary Target (Option C): Lift Curve Architecture

- Target Cl0: 0.4
- Target Cl_alpha: 0.11 /deg

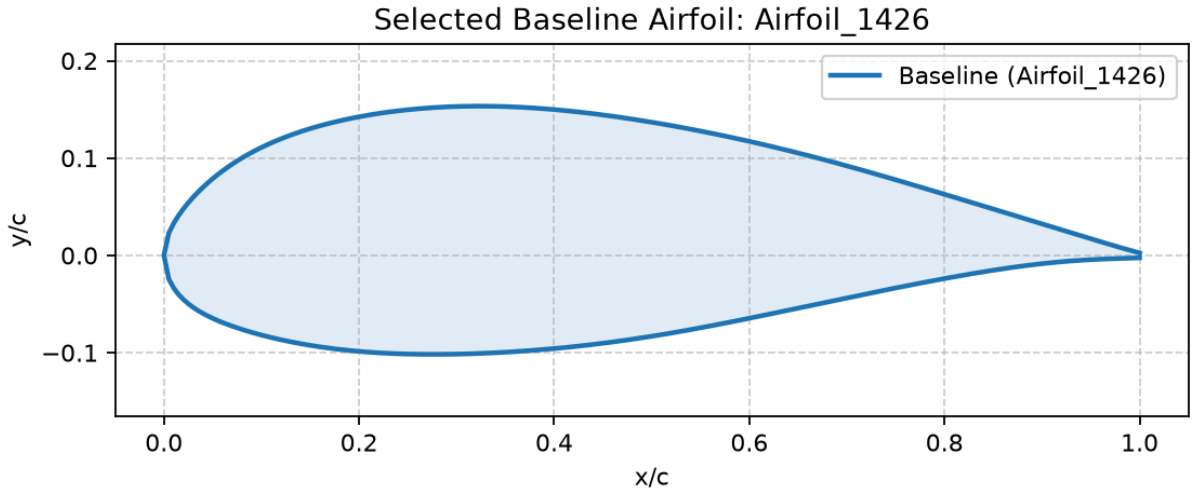
☐ Constrain Pitching Moment (Cm)

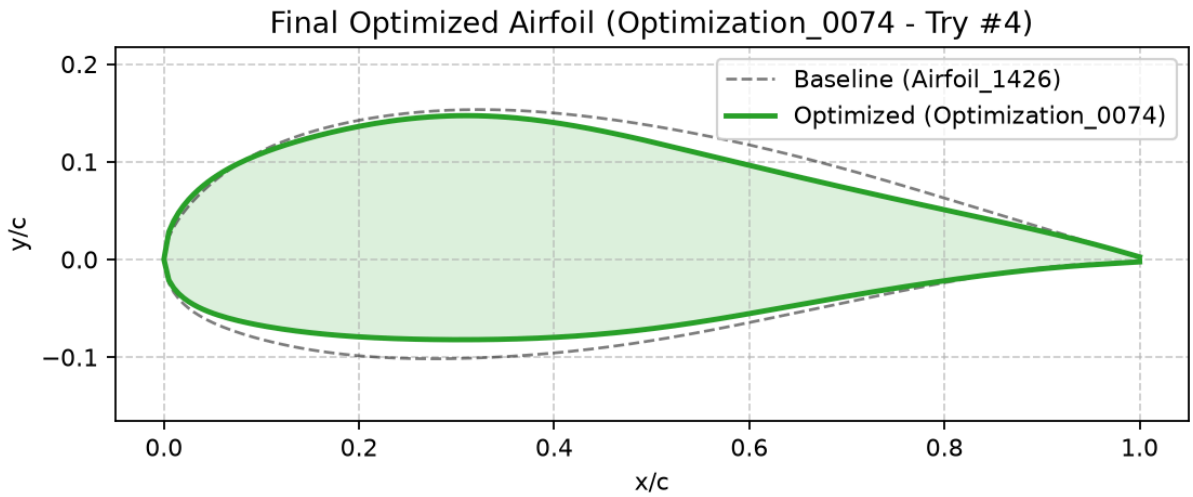
☑ Shape Stall Behavior (closed)

2. Rate of Accuracy Compared to Original User Requirements

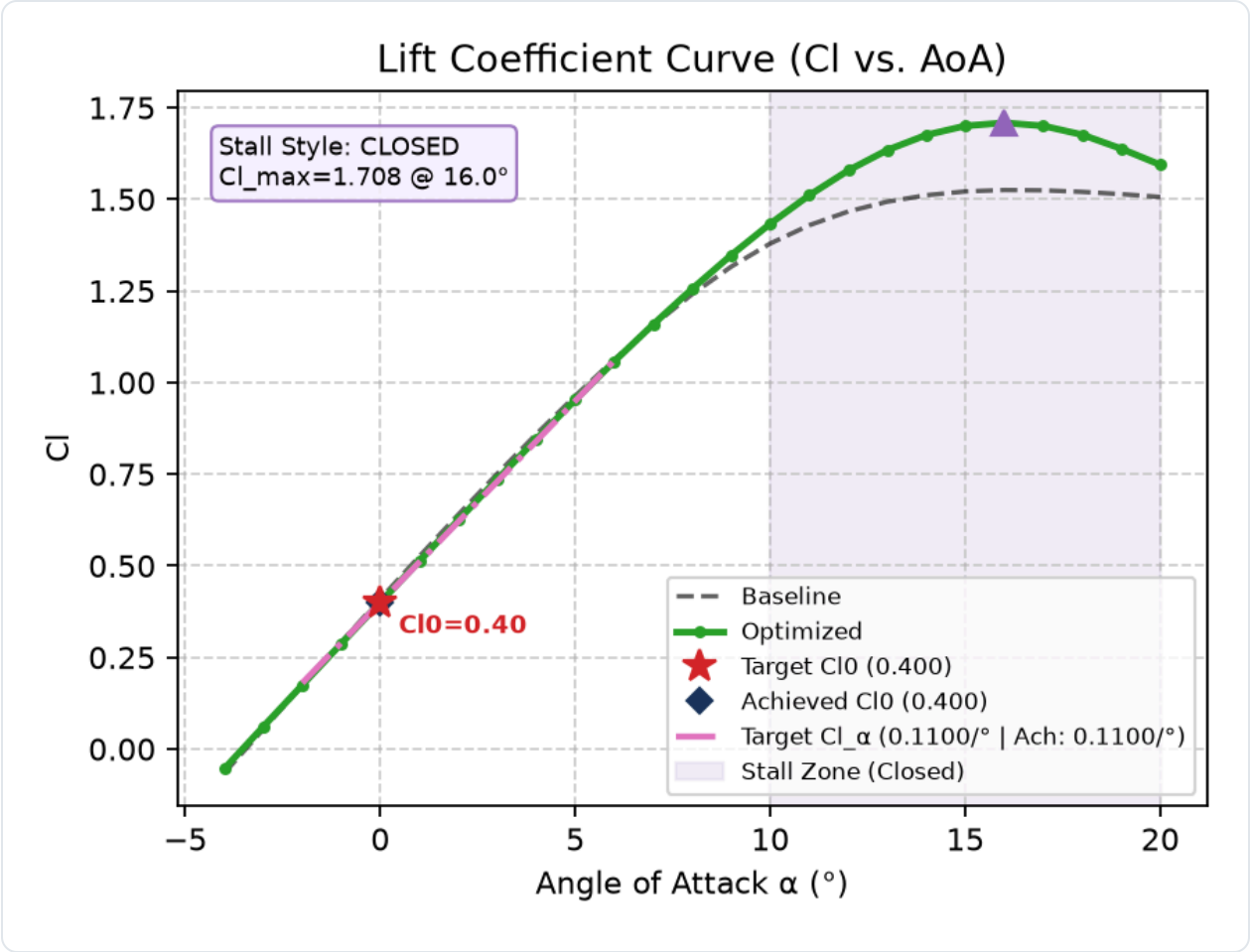
Requested Parameter	User Target	Baseline Value	Optimized Value	Absolute Error	Accuracy Rate
Zero-Angle Lift (Cl0)	0.4000	0.4094	**0.4001**	0.0001	**99.98%**
Lift Slope (Cl_alpha [/deg])	0.1100	0.1106	**0.1100**	0.0000	**99.97%**

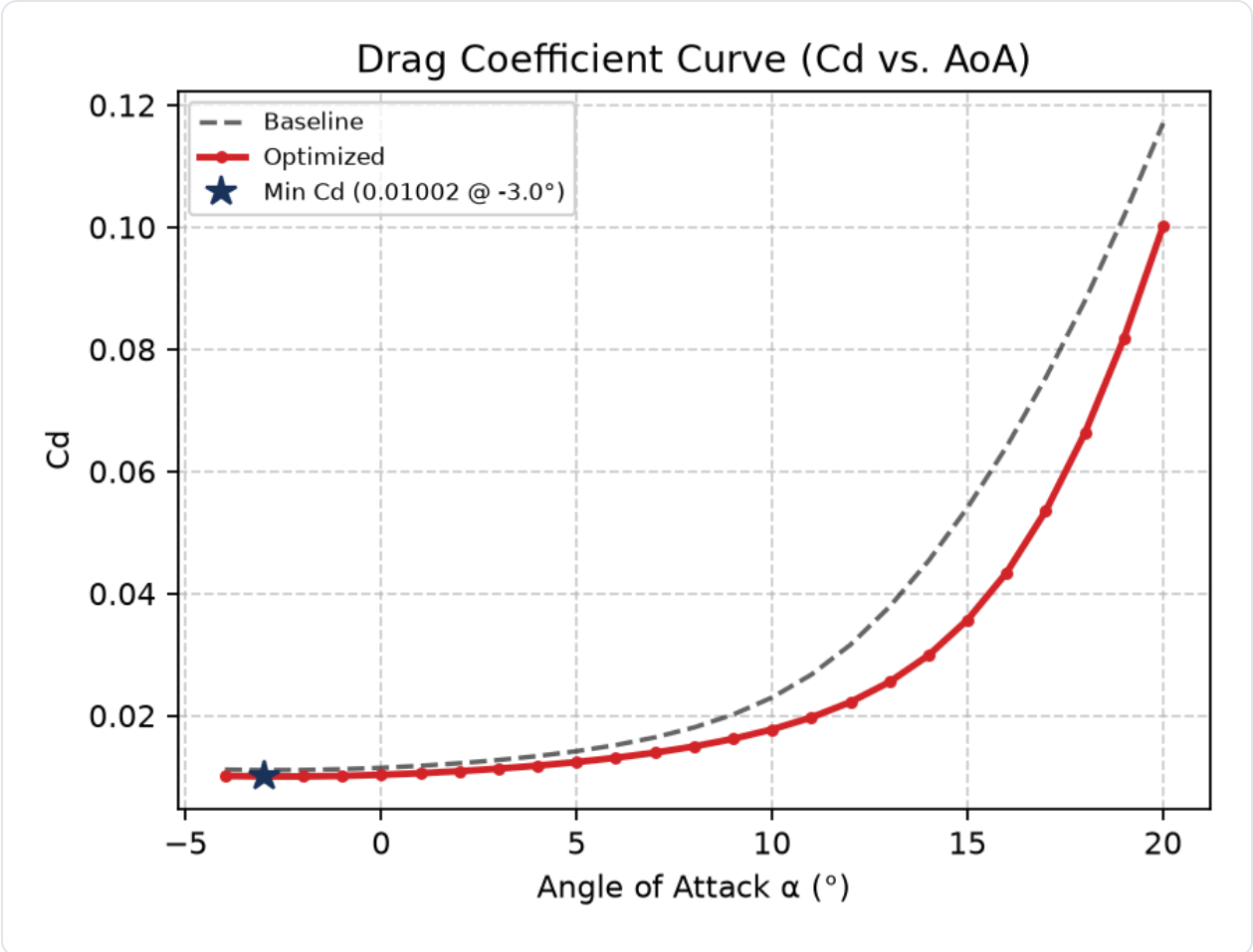
3. Baseline Airfoil vs. Final Optimized Airfoil

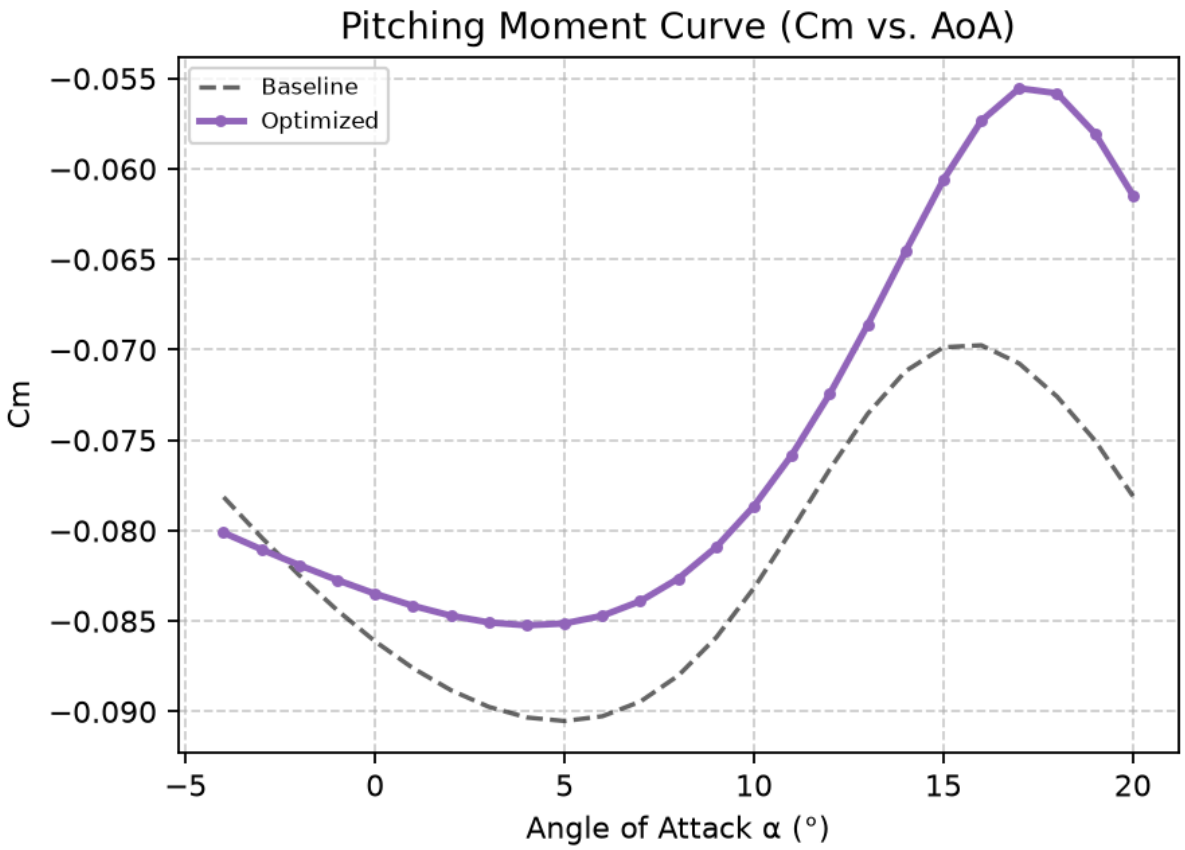




4. Aerodynamic Polar Curves (-4° to 20° AoA)


Lift Coefficient Curve (Cl vs. AoA)
Stall Style: CLOSED
Cl_max=1.708 @ 16.0°
Cl0=0.40
Baseline
Optimized
Target Cl0 (0.400)
Achieved Cl0 (0.400)
Target Cl_α (0.1100/° | Ach: 0.1100/°)
Stall Zone (Closed)
Cl
Angle of Attack α (°)


Drag Coefficient Curve (Cd vs. AoA)
Baseline
Optimized
Min Cd (0.01002 @ -3.0°)
Cd
Angle of Attack α (°)



5. Optimization Trajectory (14 Tries Evaluated)

Try	Starting Weight Modification	Initial Score	Final Score	Iterations	Time Taken
Try #01	Exact Baseline	0.30046	0.01281	6	4.17s
Try #02	Perturbed (+/-2.0%)	0.29604	0.01064	10	10.85s
Try #03	Perturbed (+/-2.5%)	0.25275	0.01100	10	5.05s
Try #04	**Perturbed (+/-3.0%)**	**0.53019**	**0.01038**	**12**	**5.95s**
Try #05	Perturbed (+/-3.4%)	0.26590	0.01083	14	7.02s
Try #06	Perturbed (+/-3.9%)	0.37090	0.01062	12	7.88s

Try #07	Perturbed (+/-4.4%)	0.25719	0.01068	11	6.41s
Try #08	Perturbed (+/-4.9%)	0.31915	0.01116	11	7.26s
Try #09	Perturbed (+/-5.4%)	0.28619	0.01045	8	5.14s
Try #10	Perturbed (+/-5.8%)	0.37434	0.01136	16	11.78s
Try #11	Perturbed (+/-6.3%)	1.17367	0.01062	11	9.57s
Try #12	Perturbed (+/-6.8%)	0.31544	0.01194	17	10.47s
Try #13	Perturbed (+/-7.3%)	0.39411	0.01065	11	7.53s
Try #14	Perturbed (+/-7.8%)	0.18659	0.01041	9	7.46s

6. 19 CST Weights (Baseline vs. Optimized)

Copy 19 CST Weights

```
-0.308922, -0.121481, -0.390755, 0.093661, -0.594904, 0.113978, -0.285222, -0.015492, -0.070627, 0.39
```

CST Weight	Baseline (Airfoil_1426)	Optimized (Optimization_0074)	Delta (Δ)
CST_00 (Lower)	-0.352246	-0.308922	+0.043324
CST_01 (Lower)	-0.161975	-0.121481	+0.040494
CST_02 (Lower)	-0.490106	-0.390755	+0.099351
CST_03 (Lower)	+0.079514	+0.093661	+0.014147
CST_04 (Lower)	-0.640715	-0.594904	+0.045811
CST_05 (Lower)	+0.121180	+0.113978	-0.007202
CST_06 (Lower)	-0.370154	-0.285222	+0.084932
CST_07 (Lower)	+0.014508	-0.015492	-0.030000

CST_08 (Lower)	-0.040627	-0.070627	-0.030000
CST_09 (Upper)	+0.317646	+0.397057	+0.079411
CST_10 (Upper)	+0.523901	+0.392926	-0.130975
CST_11 (Upper)	+0.186018	+0.232451	+0.046433
CST_12 (Upper)	+0.713066	+0.730156	+0.017090
CST_13 (Upper)	+0.043593	+0.013593	-0.030000
CST_14 (Upper)	+0.651777	+0.488832	-0.162944
CST_15 (Upper)	+0.198252	+0.152645	-0.045607
CST_16 (Upper)	+0.393377	+0.295033	-0.098344
CST_17 (Upper)	+0.268768	+0.294894	+0.026126
CST_18 (TE_Gap)	+0.005000	+0.005000	-0.000000

7. Numerical Aerodynamic Coefficients (-4.0° to 20.0° AoA)

AoA (°)	Lift Coeff (Cl)	Drag Coeff (Cd)	Moment Coeff (Cm)	Glide Ratio (L/D)
-4.0°	-0.05365	0.01011	-0.08011	-5.31
-3.0°	+0.06113	0.01002	-0.08104	6.10
-2.0°	+0.17471	0.01002	-0.08191	17.43
-1.0°	+0.28753	0.01011	-0.08273	28.43
+0.0°	+0.40006	0.01029	-0.08349	38.88
+1.0°	+0.51250	0.01055	-0.08416	48.59
+2.0°	+0.62461	0.01088	-0.08470	57.40
+3.0°	+0.73582	0.01129	-0.08508	65.15
+4.0°	+0.84546	0.01179	-0.08524	71.72
+5.0°	+0.95290	0.01238	-0.08513	77.00
+6.0°	+1.05756	0.01307	-0.08470	80.89
+7.0°	+1.15881	0.01391	-0.08390	83.30

+8.0°	+1.25589	0.01493	-0.08265	84.13
+9.0°	+1.34780	0.01617	-0.08091	83.34
+10.0°	+1.43332	0.01772	-0.07864	80.89
+11.0°	+1.51093	0.01967	-0.07580	76.80
+12.0°	+1.57868	0.02219	-0.07244	71.15
+13.0°	+1.63424	0.02548	-0.06862	64.13
+14.0°	+1.67525	0.02986	-0.06455	56.10
+15.0°	+1.70018	0.03569	-0.06060	47.63
+16.0°	+1.70840	0.04342	-0.05734	39.34
+17.0°	+1.70014	0.05354	-0.05554	31.76
+18.0°	+1.67598	0.06628	-0.05582	25.29
+19.0°	+1.63777	0.08181	-0.05805	20.02
+20.0°	+1.59400	0.10014	-0.06145	15.92

8. 400 CAD Coordinates for SolidWorks (X, Y, Z=0)

Copy 400 SolidWorks Coordinates

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1.000000        0.002500        0.000000
0.994975        0.003965        0.000000
0.989950        0.005420        0.000000
0.984925        0.006863        0.000000
0.979899        0.008291        0.000000
0.974874        0.009704        0.000000
0.969849        0.011099        0.000000
0.964824        0.012478        0.000000
0.959799        0.013838        0.000000
0.954774        0.015180        0.000000
0.949749        0.016504        0.000000
0.944724        0.017810        0.000000
0.939698        0.019098        0.000000
0.934673        0.020369        0.000000
0.929648        0.021623        0.000000
0.924623        0.022862        0.000000
0.919598        0.024085        0.000000
0.914573        0.025294        0.000000
0.909548        0.026489        0.000000
0.904523        0.027671        0.000000
0.899497        0.028842        0.000000
0.894472        0.030002        0.000000
0.889447        0.031152        0.000000
0.884422        0.032292        0.000000
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